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1 message

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5.3 Astrophysical Evidence: Black Holes as a Source of Dark Energy (Farrah et al., ApJL, 2023)

In 2023, a team led by Duncan Farrah published an open-access paper in The Astrophysical Journal Letters titled "Observational Evidence for Cosmological Coupling of Black Holes and its Implications for an Astrophysical Source of Dark Energy". The authors analysed the masses of black holes across cosmic history and found that they grow in a way that is "cosmologically coupled". Zero cosmological coupling is excluded at 99.98% confidence. They conclude that black holes themselves provide an astrophysical source of dark energy – the very force that accelerates the expansion of the universe.

How this aligns with the Harmonic Framework (GUT):

- The GUT proposes that the paired potential m' (the reverse-time mass component) is the physical origin of dark energy.
- Black holes, as regions where $n \rightarrow 0$ (phase nodes), couple directly to the Tau lattice (the 27 Hz anchor). Their observed mass growth is not due to accretion of ordinary matter, but to the conversion of paired potential energy ($m + m' = 0$) into measurable mass.
- The net repulsive effect at cosmic scales is exactly what Farrah et al. measure as "dark energy".

What the GUT adds that the paper lacks:

- The paper does not cite my 2018 work or the Harmonic Framework. It provides no specific physical mechanism for the coupling – only a statistical inference.
- The GUT supplies the missing mechanism: the 27 Hz Tau lattice, the harmonic ladder ($n = \ln(P)/\ln(27)$), and the paired potential condition $m + m' = 0$.
- The GUT also makes testable predictions (27 Hz gravitational wave line, galaxy spin bias, harmonic overtones in black hole mergers) that go beyond the statistical correlation of Farrah et al.

Significance for the GUT's empirical proof:

This paper is independent, peer-reviewed, observational confirmation that black holes are linked to a universal field that drives cosmic expansion. The GUT predicted this link (via m' and the Tau lattice) years before the paper was published. Therefore, Farrah et al. (2023) provides independent astrophysical support for the core principle of the Harmonic Framework, even though they do not cite it.

Citation:

Farrah, D., Croker, K.S., Zevin, M., et al. 2023, The Astrophysical Journal Letters, 944, L31.
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